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Sign Language Recognition: A Large-scale Multi-view Dataset and Comprehensive Evaluation

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Abstract

Vision-based sign language recognition is an extensively researched problem aimed at advancing communication between deaf and hearing individuals. Numerous Sign Language Recognition (SLR) datasets have been introduced to promote research in this field, spanning multiple languages, vocabulary sizes, and signers. However, most existing popular datasets focus predominantly on the frontal view of signers, neglecting visual information from other perspectives. In practice, many sign languages contain words that have similar hand movements and expressions, making it challenging to differentiate between them from a single frontal view. Although a few studies have proposed sign language datasets using multi-view data, these datasets remain limited in vocabulary size and scale, hindering their generalizability and practicality. To address this issue, we introduce a new large-scale, multi-view sign language recognition dataset spanning 1,000 glosses and 30 signers, resulting in over 84,000 multi-view videos. To the best of our knowledge, this is the first multi-view sign language recognition dataset of this scale. In conjunction with offering a comprehensive dataset, we perform extensive experiments to assess the performance of state-of-the-art Sign Language Recognition models utilizing on our dataset. The findings indicate that utilizing multi-view data substantially enhances model accuracy across all models, with a maximum performance improvement of up to 19.75% compared to models trained on single-view data. Our dataset and baseline models are publicly accessible on GitHub[1].

1. Introduction

Sign language is the primary mode of communication within the deaf community and serves as a crucial bridge that allows deaf individuals to communicate with both each other and hearing people. However, understanding sign language can be difficult, and as a result, relatively few hearing individuals are proficient in it [3]. This communication gap has driven the development of sign language translation tools, which rely on advanced sign language recognition models as their core technology [11]. These tools aim to improve communication and understanding between deaf and hearing individuals by translating sign language into spoken or written forms.

Sign language recognition methods are broadly classified into two main categories: word-level recognition (i.e., isolated sign language recognition) [13, 33, 35, 57], and sentence-level recognition (i.e., continuous sign language recognition) [21, 44, 66]. This study focuses on word-level recognition, as it serves as a fundamental building block for more advanced tasks in this domain. Word-level recognition datasets have garnered considerable interest from the research community and have been extensively studied over the years [4, 32, 58, 64]. Most existing studies predominantly focus on recognition from a frontal view perspective. However, our findings indicate that restricting recognition to frontal view data can significantly reduce model accuracy for two main reasons. First, in practical applications, sign language data is often captured from various angles, not exclusively from a frontal perspective. Consequently, training models solely on frontal view data may not yield optimal performance when applied to diverse, real-world input angles, as shown in Table 1. Second, many signs appear visually similar when viewed from a frontal angle, making it difficult for models trained on such data to accurately differentiate between them (see Figure 1).

Incorporating multiple viewing angles into the input data is crucial for enhancing the accuracy and robustness of sign language recognition models. Accordingly, advancing the accuracy of sign language recognition necessitates the integration of multi-view video data. While some early studies have explored the use of multiple camera views for sign language recognition [16, 53], they have been limited

[1]Available at https://github.com/Etdihatthoc/Multi-VSL_WACV_2025

Table 1. **Performance degradation of the models trained on front-view data when tested with other views.** The *(numbers)* in parentheses indicate the extent of performance degradation.

Test view	I3D [57]		Swin Tr. [35]		VTNPF [13]	
	Top-1	*Top-5*	*Top-1*	*Top-5*	*Top-1*	*Top-5*
Frontal	**40.60**	**73.03**	**76.43**	**93.59**	**66.21**	**90.11**
Right	7.84 *(-32.76)*	20.81 *(-52.22)*	29.57 *(-46.86)*	53.35 *(-40.24)*	19.31 *(-46.90)*	41.09 *(-49.02)*
Left	5.09 *(-35.51)*	14.52 *(-58.51)*	18.40 *(-58.03)*	37.04 *(-56.55)*	11.79 *(-54.42)*	27.83 *(-62.28)*

by small and homogeneous datasets, typically featuring a narrow range of vocabulary (50 glosses[2] in [16] and 200 glosses in [53]) and involving only a small number of signers (14 signers in [16] and 5 signers in [53]). These limitations have constrained the generalizability and broader applicability of their results.

To bridge this gap, our study is the first to introduce a large-scale dataset specifically tailored for multi-view sign language recognition. Furthermore, we perform thorough empirical studies to evaluate the effectiveness of using multi-view data in enhancing recognition accuracy and to benchmark the performance of state-of-the-art models using this comprehensive dataset. Our approach aims to highlight the benefits of multi-view datasets in enhancing the reliability and accuracy of sign language recognition systems, ultimately providing a valuable foundation for future research in this field. Our key contributions are as follows:

- We develop a new dataset for multi-view Vietnamese sign language recognition that includes 1,000 glosses, over 84,000 videos and involves 30 different signers. As far as we know, this is the first multi-view sign language dataset available for this language and the largest of its kind across all languages. This dataset is expected to serve as a crucial resource for advancing research and development in the field. (see Figure 2).
- We perform extensive experiments using state-of-the-art models to thoroughly evaluate the benefits of incorporating multiple views, investigate the characteristics of different views, and assess the performance of various recognition methods. Our experimental findings reveal that utilizing multiple views substantially enhances recognition accuracy, highlighting the effectiveness of this approach across all the models.
- We propose a straightforward method to tackle the issue of missing modalities (situations where some views are unavailable in real-world scenarios). Our approach shows the advantages of utilizing multi-views during training, even when complete views are not available during the testing phase.

[2]A gloss represents the fundamental lexical unit used for annotating sign languages.

The paper is organized as follows. Section 2 reviews key research on Sign Language Recognition, covering popular SLR datasets (2.1) and recognition methods (2.2). Afterward, Section 3 introduces our new large-scale multi-view SLR dataset, detailing its creation process and features. Section 4.1 presents experimental results and analysis, highlighting the superior performance of multi-view Sign Language Recognition over traditional single-view methods. We conclude the paper in Section 5.

2. Related work

In this section, we provide a brief overview of existing sign language datasets and state-of-the-art sign language recognition methods to highlight the need for a large-scale multi-view Multi-VSL dataset.

2.1. Sign language recognition datasets

The field of sign language recognition (SLR) has greatly benefited from the development of diverse datasets, which provide the foundation for training and evaluating automated recognition systems. Several word-level hand gesture sign language (HGS) datasets have been particularly influential in advancing the state of the art. The MS-ASL [27] and WLASL [32] datasets are the most popular large-scale datasets for American Sign Language, containing 1,000 and 2,000 commonly used ASL signs, respectively. Other notable ASL datasets include ASL Citizen [14], a crowdsourced collection of 2,731 signs from 52 signers in real-world settings, and PopSign ASL [52], gathered via a smartphone-based educational game to help hearing parents learn and practice ASL with their deaf children. Large datasets in other languages include Turkish [49], Spanish [47], German [39], Chinese [10], Argentinian [45], and Russian [28].

Additionally, the KL_MV2DSL dataset [53] provides a comprehensive multi-view perspective, featuring 200 Indian Sign Language signs captured from five distinct angles, resulting in a diverse set of 2,500 video samples. The MVSL dataset [16] further enriches the SLR field by offering 10,500 high-resolution videos of 50 Chinese sign language words, recorded from three different viewpoints, performed by 14 signers. These datasets, varying in language, signers, and capture modalities, collectively enhance the robustness and accuracy of SLR models, facilitating more nuanced and comprehensive recognition of sign language gestures.

Previous works have introduced large-scale and diverse datasets featuring numerous glosses and signers but these resources predominantly capture only the frontal view of the signers or a small number of multi-view instances. Meanwhile, in computer vision, large-scale multi-view video datasets are key for training models to understand complex human behaviors from different angles. The Human3.6M

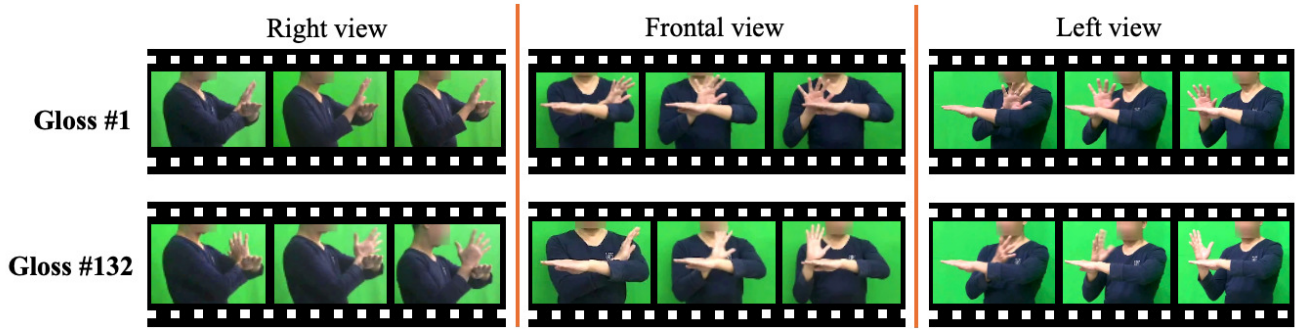


Figure 1. A signer performs gloss #1 and #132, which are two words with quite similar gestures. The biggest difference is that gloss #132 includes a hand-shaking motion when moving from left to right, while gloss #1 keeps the hand still. This difference is more clearly visible from the Right and Left views rather than just using the Frontal view.

Table 2. A comparison between our proposed dataset and existing Sign Language Recognition datasets.

Year	Dataset	Glosses	Videos	Signers	Language	Type
2016	LSE-sign [47]	2,400	2,400	2	Spanish	Single-View
2016	DEVISIGN [10]	2,000	24,000	8	Chinese	Single-View
2018	MS-ASL [27]	1,000	25,513	222	American	Single-View
2019	DGS Kinect [39]	40	3,000	15	German	Single-View
2020	AUTSL [49]	226	38,336	43	Turkish	Single-View
2020	WLASL [32]	2,000	21,083	119	American	Single-View
2023	LSA64 [45]	64	3,200	10	Argentinian	Single-View
2023	Slovo [28]	1,000	20,000	194	Russian	Single-View
2023	ASL Ciztizen [14]	2,731	83,399	52	American	Single-View
2023	PopSign ASL [52]	250	214,326	47	American	Single-View
2021	KL_MV2DSL [53]	200	2,500	5	Indian	Multi-view
2022	MVSL [16]	50	10,500	14	Chinese	Multi-view
2024	**Multi-VSL (Ours)**	**1,000**	**84,764**	**30**	**Vietnamese**	**Multi-view**

dataset [24] captures 11 actors performing 17 actions using four cameras, enhancing virtual and augmented reality systems. Similarly, the NTU RGB+D dataset [47] uses three cameras at different angles to improve action recognition for applications like security surveillance. Carnegie Mellon's Panoptic Studio [26] pushes further, using over 500 cameras to study social interactions, advancing group behavior analysis.

Singular perspective datasets often fail to encompass the complex array of body and hand movements integral to sign language, potentially compromising recognition accuracy. Furthermore, current gloss-level sign recognition models are primarily trained and evaluated on either proprietary datasets [17, 22, 31, 63] or small-scale datasets comprising fewer than one hundred words [22, 31, 34, 38, 43, 59, 63]. Such datasets limit model generalization as they do not fully capture the complexity and diversity of sign languages in real-world settings.

2.2. Sign Language Recognition Models

Sign Language Recognition (SLR) has seen extensive research over the years. Early approaches typically utilize handcrafted image features, such as SIFT [54, 61] or HOG [6,29,36] or DCT [1], to extract visual representations from RGB frames of sign language gestures. These features were typically followed by classification algorithms like SVM [36, 40, 41], Hidden Markov Models [1, 29, 51], Gradient Boosted Trees [42] for classifying glosses.

With the rise of deep learning and the availability of large-scale datasets, more sophisticated deep learning-based sign language recognition models have emerged. Many methods operate directly on the RGB frames of the sign language recognition videos, typically utilizing CNN-based architectures [12, 32, 48] as visual feature extractors, combined with specialized layers like RNNs [12, 15, 62], or Transformers [7] to learn the temporal patterns of sign language videos. Instead of modeling the spatial and temporal information separately, many researchers have proposed

jointly learning spatio-temporal features with 3D-CNN architectures [23, 57]. Newer works have also adopted Vision Transformers [8, 30, 33, 35] as the image feature extractor for enhanced representation.

However, visual-based methods might suffer from noisy background signals, distracting the information from the signers' hand gestures and poses. To mitigate this problem, many works have extracted skeletal data using frameworks like mmPose [46], MediaPipe [37] to be used as input for the recognition model. Convolutional layers have been applied to capture the spatio-temporal dynamic of human pose sequences [2, 9, 13]. More recent efforts [18–20, 55, 60] employ graph convolutional networks [65] (GCNs) to model the human pose's representations, thanks to its well-structured nature, and displayed promising results. In addition, many multimodal approaches have been proposed to combine diverse features from RGB videos and skeletal data [5, 13, 18–20], or other modalities such as optical flow [25], further improving recognition accuracy.

Although advanced deep learning models for sign language recognition have shown promising results, their performance on large-scale, real-world datasets remains suboptimal [27, 32]. A significant contributing factor is the limited visual representation due to relying on a single frontal view, or only a small amount of diverse multi-view data. This restriction makes it difficult to differentiate between visually similar signs, especially when dealing with extensive sign vocabularies. To address this limitation, we introduce a new dataset with three views, along with a large number of videos and glosses, to better reflect real-world conditions and enhance the robustness of sign language recognition models. Our dataset aims to provide a more comprehensive training and evaluation resource that can accommodate varying input orientations and improve the generalizability of models in diverse environments. This approach will help bridge the gap between current research and practical applications, fostering more effective and inclusive communication technologies for the deaf and hard-of-hearing communities. Table 2 summarizes popular existing sign language datasets and compares them with Multi-VSL, our proposed large-scale multi-view dataset.

3. Multi-VSL: A Large-scale Multi-view Sign Language Dataset

In this section, we start by describing the data sources and collection methods, including processing, annotation, de-identification, etc., and then provide detailed statistics of our dataset.

3.1. Dataset Creation Process

A key feature of this dataset is the use of a three-camera setup, recording glosses from various angles to provide multidimensional and rich visual data. An overview diagram of the whole process is presented in Figure 3, which includes four main steps: video data collection, video standardization, gloss splitting and annotation, followed by de-identification. Each step is executed to optimize data quality, thereby enhancing the effectiveness of SLR models.



Figure 2. Diverse signers perform different glosses in three separate views left, front, and right.

3.1.1 Video data collection

To construct our comprehensive three-view sign language dataset, we collaborate with signers from Hanoi Nhan Chinh School for the Deaf to ensure expertise and specificity in the sign language recognition (SLR) dataset. Each participant is tasked with recording a series of glosses derived directly from Vietnamese Sign Language Dictionary ensuring a wide representation of the language's vocabulary. This approach is designed to capture a rich diversity within the dataset, which is crucial for minimizing the risk of overfitting between the glosses and signers and enhancing the generalizability of our models.

To maximize efficiency during the recording process, signers complete the recording of several hundred glosses in a single session, which are strategically divided into four distinct segments based on gloss IDs: 1-200, 200-400, 400-700, and 700-1000. This structuring not only facilitates systematic recording but also helps manage the voluminous data more effectively. For each shot, a slide containing all the videos of the glosses is displayed on a large screen behind the camera, helping the signers perform the words correctly. The audio in the videos assists with annotation.

A key innovation in our data collection method is the use of three separate cameras, each positioned at different strategic angles: Left ($\sim$45°), Front ($\sim$0°), and Right ($\sim$45°). This tri-camera setup, relatively rare in existing

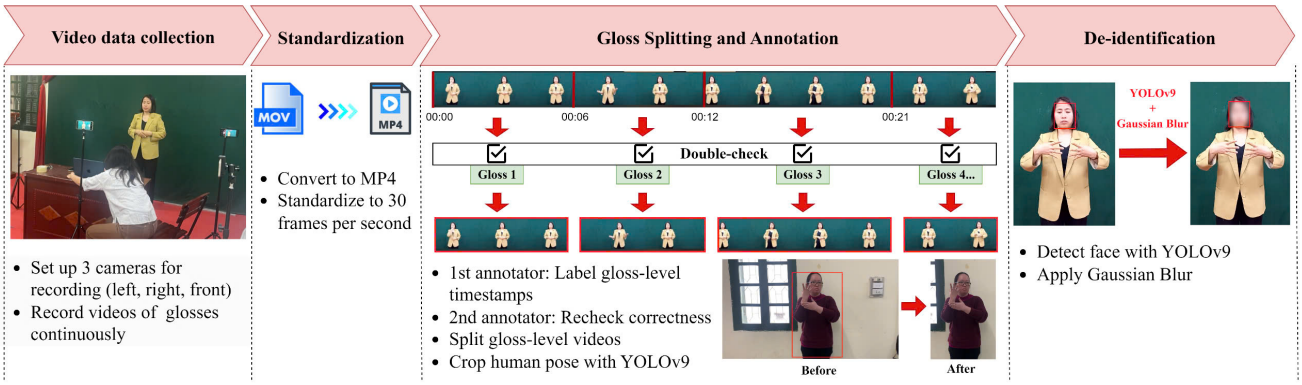


Figure 3. **Illustration of our dataset creation pipeline.** There are four main steps: Data collection, Standardization, Gloss Splitting and Annotation, and De-identification.

word-level Sign Language Recognition datasets, provides a multifaceted view of each sign. The diverse perspectives captured by this setup greatly enhance the utility of the dataset for various machine learning and computer vision research applications, enabling more accurate and robust recognition algorithms.

3.1.2 Video Standardization

The collected videos come in various formats, such as MOV from iPhones, AVI from multiple smartphones, and formats from personal camcorders. To ensure compatibility and streamline processing, all videos are converted to a standardized MP4 format. Since the frame rates vary significantly due to the use of different recording devices, this inconsistency poses challenges for processing and machine learning tasks that require uniform data. To address this, we standardize all videos to a consistent frame rate of 30 frames per second, ensuring uniformity for analysis and model training.

3.1.3 Gloss Splitting and Annotation

After standardization, videos are uploaded to our in-house annotation platform. For each video, we employ two annotators: the first splits the original video into smaller portions corresponding to the glosses, while the second performs a verification to ensure accuracy. Additionally, we utilize YOLOv9 [56] for automated human pose detection and cropping, which helps reduce background distractions and enhances the overall quality of the dataset.

3.1.4 De-identification

Due to stringent requirements for privacy and anonymity, we employ a pretrained YOLOv9 model [50] for facial detection coupled with Gaussian blurring techniques to obscure faces in the videos. This step ensures that while the expressive details necessary for sign language recognition remain clear, the identity of the participants is protected.

Table 3. Characteristics of our three subsets: **Multi-VSL 200**, **Multi-VSL 400** and **Multi-VSL 1000**.

Characteristic	M-VSL 200	M-VSL 400	M-VSL 1000
Words	199	400	1,000
Signers	28	30	30
Videos (all views)	18,981	35,147	84,764
Instances (multi-view)	5,663	11,637	28,176
Avg. instances/gloss	28.46	29.09	28.18
Train/Val/Test (signers)	20/4/4	21/4/5	21/4/5

3.2. Dataset Characteristics

After the data curation process, we obtained the final dataset consisting of 84,764 word-level videos, recorded by 30 signers, with a vocabulary of 1,000 unique words selected from Vietnamese Sign Language Dictionary. Every video captures a single signer performing one gloss from one out of three viewpoints (left, front, right), with an average duration of approximately 2.67 seconds. In addition, among the 1000 glosses chosen to build the dataset, we deliberately select some words with similar sign movements in sign language as Figure 1 to increase the challenge and diversity of the dataset.

Following practices from [32], we create three subsets from the full 1000-gloss Multi-VSL dataset, each with different vocabulary sizes, to examine the impact of vocabulary size on the performance of the sign language recognition model. Specifically, we select the top K glosses, where $K = \{200, 400, 1000\}$, and arrange them into three subsets: **Multi-VSL 200**, **Multi-VSL 400**, and **Multi-VSL 1000** (the full dataset). The train, validation, and test splits of these subsets are determined based on the signer, ensuring that the data is partitioned according to the individuals performing the signs. The characteristics of these subsets are reported in Table 3.

Table 4. The baseline models used in the experiments.

Category	Baseline Model	Year Released
RGB-based	I3D [57]	2019
	Swin Trans [35]	2022
	MVitV2 [33]	2022
Pose flow-based	VTNHCPF [13]	2021

4. Benchmark Evaluation

In this section, we perform comprehensive experiments to evaluate the performance of existing methods on our dataset and examine the effectiveness of incorporating multi-view approaches.

4.1. Experiment Setup

Research Questions. We investigate the following research questions:

1. **Multi-view vs. Single-view**: We assess the degree to which recognition accuracy can be enhanced by employing multiple views, in contrast to the single-view approach typically used in conventional methods.
2. **Effect of Vocabulary Size**: We evaluate the impact of vocabulary size on recognition accuracy by comparing performance across datasets with varying numbers of classes.
3. **Robustness to Missing Views**: In practical scenarios, the data may not encompass all possible views. Therefore, we investigate the effect of missing or distorted views on recognition accuracy to assess the model's robustness under real-world conditions.
4. **Effectiveness towards ambiguous cases**: We present the most commonly confused pairs of glosses and qualitatively demonstrate how utilizing multiple views enhances the recognition capabilities of difficult, similar glosses compared to a single-view approach.

Data Pre-processing and Augmentation. All video frames in our dataset are resized to a resolution of (320 $\times$ 256) and converted to the RGB format. Each video is divided into 16 segments of equal length, and one frame is randomly sampled from each of the 16 segments to create a sequence of 16 frames, which serve as input for the predictive models. For Poseflow-based methods, we utilize mmPose [46] to extract skeletal key points.

To diversify the original dataset and prevent overfitting, we subject the training videos to a random combination of various data augmentation techniques: random cropping, horizontal flipping, rotation, shearing, salt-and-pepper noise, and adjustments to brightness, contrast, saturation, and hue.

Benchmarks. To evaluate the performance of our datasets, we choose recent popular, state-of-the-art Action Recognition/Sign Language Recognition models. We also pick representatives from both RGB-based and pose-based methods to ensure a comprehensive evaluation across different types of visual data. Specifically, we select three RGB baselines: 3D-CNN based I3D [57], Video Swin Transformer [35] and MVitV2 [33] and one pose-based model: VTNHCPF [13] .

Multi-view SLR framework The chosen benchmarks are only designed to handle a singular view input, whether it is RGB frames or poses. Given a single input X, which can be RGB vectors or pose vectors, these models employ an encoder model E to extract the latent features Z from the input, before applying a classifier layer on the latent representation Z.

To adapt these models effectively for our three-view data X_f, X_r, X_l, representing the front, right and left views respectively, we initialize three identical encoders E_f, E_r, E_l, with each encoder extracting specific features of a view.

$$Z_f = E_f(X_f), \quad Z_l = E_l(X_l), \quad Z_r = E_r(X_r).$$

We then concatenate the three view-specific latent features Z_f, Z_l, Z_r to combine the representation of the three visual views and apply a final classifier layer, a Multilayer Perceptron, for sign language recognition

$$y = MLP\left([Z_f, Z_l, Z_r]\right).$$

Evaluation Metrics. We evaluate the performance of the baseline models on three subdatasets: Multi-VSL 200, Multi-VSL 400, and Multi-VSL 1000. We report the average top-K classification accuracy with K being 1 and 5, over all glosses. In sign languages, certain signs might have similar gestures ,which can lead to misclassifications, but additional context from top model predictions can help resolve these errors. Thus, using top-K predicted labels is more effective for word-level sign language recognition.

4.2. Multi-view vs. Single-view

Table 5 presents the performance of Sign Language Recognition baselines on our dataset, evaluated in both single-view and three-view settings.

Among all baselines, the three-view models exhibit substantial improvements in accuracy over their single-view counterparts. Most notably, the I3D three-view model achieves an improvement of 19.75%, and VTNPF obtains a performance gain of 10.11% compared to their respective single-view versions. Another observation is that the improvements of three-view models are more significant for the top-1 accuracy compared to the top-5 accuracy metrics. We hypothesize that this advantage is due to the three-view representation's ability to distinguish between commonly confused words. These findings demonstrate the superior representation capabilities of multi-view data compared to the traditional single-view approach, allowing for a more comprehensive capture of the nuances in sign language glosses, and ultimately leading to improved recognition performance.

Table 5. **Comparison of single-view and multi-view approaches across various benchmarks.** The experiments are conducted on three datasets containing 200, 400, and 1000 glosses, respectively. The rows labeled *Imprv.* show the improvement margins of the 3-view approach compared to the 1-view approach.

Model	Num of views	Multi-VSL 200		Multi-VSL 400		Multi-VSL 1000	
		Top-1 Acc.	*Top-5 Acc.*	*Top-1 Acc.*	*Top-5 Acc.*	*Top-1 Acc.*	*Top-5 Acc.*
I3D [57]	1-view	56.62 ± 0.0055	83.99 ± 0.0063	52.84 ± 0.0063	84.42 ± 0.0045	40.60 ± 0.0040	73.03 ± 0.0046
	3-view	66.35 ± 0.0066	90.14 ± 0.0050	63.14 ± 0.0039	88.48 ± 0.0025	60.35 ± 0.0055	86.52 ± 0.0021
	Imprv.	*+9.73*	*+6.15*	*+10.30*	*+4.06*	*+19.75*	*+13.49*
Swin Trans. [35]	1-view	77.29 ± 0.0083	91.81 ± 0.0064	77.75 ± 0.0064	93.41 ± 0.0016	76.43 ± 0.0045	93.59 ± 0.0014
	3-view	82.33 ± 0.0034	94.03 ± 0.0032	79.93 ± 0.0020	93.79 ± 0.0022	76.73 ± 0.0031	93.83 ± 0.0004
	Imprv.	*+5.04*	*+2.22*	*+2.18*	*+0.38*	*+0.3*	*+0.24*
MVITv2 [33]	1-view	81.57 ± 0.0038	92.36 ± 0.0029	82.41 ± 0.0037	95.13 ± 0.0012	81.52 ± 0.0033	95.49 ± 0.0022
	3-view	86.45 ± 0.0026	95.55 ± 0.0031	85.29 ± 0.0055	96.43 ± 0.0010	84.71 ± 0.0022	96.79 ± 0.0007
	Imprv.	*+4.88*	*+3.19*	*+2.88*	*+1.30*	*+3.19*	*+1.30*
VTNPF [13]	1-view	81.64 ± 0.0052	92.29 ± 0.0037	80.67 ± 0.0039	94.89 ± 0.0029	66.21 ± 0.0027	90.11 ± 0.0017
	3-view	87.99 ± 0.0051	95.50 ± 0.0017	86.62 ± 0.0045	96.35 ± 0.0007	76.32 ± 0.0034	94.08 ± 0.0010
	Imprv.	*+6.35*	*+3.21*	*+5.95*	*+1.46*	*+10.11*	*+3.97*

Transformer-based models (*Swin Transformer, MViTv2, VTNPF)* significantly outperform the Convolution-based I3D baseline in both one-view and three-view setups. Furthermore, VTNPF, the method incorporating both RGB images and poses, is consistently the best-performing baseline across all metrics and settings. This demonstrates the effectiveness of incorporating both RGB and pose-based inputs.

4.3. Effect of vocabulary size

In this section, we assess the impact of vocabulary size on the performance of I3D and VTNPF models using our subdatasets, as shown in Table 6. Both models achieve the highest accuracy when trained and tested on the same dataset. However, as the vocabulary size increases from 200 to 1000, there is a noticeable decline in performance for both models. For instance, when trained on different datasets and tested with the Multi-VSL 200 dataset, VTNPF's accuracy decreases from 87.99% on the Multi-VSL 200 to 84.6% on the Multi-VSL 400, and further to 72.49% on the Multi-VSL 1000 dataset. This suggests that while VTNPF benefits from training on smaller datasets, the increased complexity of larger vocabularies makes it more challenging for the model to generalize effectively.

Moreover, when the models are trained on a larger dataset, such as Multi-VSL 1000, and tested on smaller datasets, both I3D and VTNPF exhibit relative stability in their accuracy, although there are differences in how well they maintain their performance. For example, when trained on the Multi-VSL 1000, I3D achieves an accuracy of 64.65% when tested on the Multi-VSL 200, 59.92% on the Multi-VSL 400, and 60.35% on the Multi-VSL 1000. These results indicate that training on a larger dataset allows both models to maintain a certain level of stability in accuracy when tested on different smaller datasets.

4.4. Robustness to missing views

In real deployment scenarios, obtaining data from all three views might not be possible for every instance. Thus, it is necessary to evaluate the robustness of three-view models trained on our dataset under settings where certain views might be missing. To simulate missing views during training/testing processes, for each data instance, we randomly delete an image from a certain view with varying missing rates (10, 25, and 50%), and replace them with zero vectors. We run the experiments with the I3D three-view model under two settings: **Full training** means the model is trained with no missing views, and tested with different missing rates; while **Partial training** means the model is trained and tested with the same missing rate. We report the results in Table 7. The experiment reveals a sharp decline in model accuracy as more frames from different views are missing. Notably, Top-1 accuracy drops from 60.35% to 43.12%, a 17.23% gap when tested with a 50% missing rate for models trained with full views.

We observe that introducing missing views during training improves robustness to missing views during testing. At a 10% missing rate, models trained with missing views show only minor drops in accuracy (1.18% for top-1, 0.05% for top-5), compared to larger declines for the full-view model (4.9% top-1, 10.44% top-5). At 50% missing, accuracy drops less sharply, from 60.35% to 57.04%.

Notably, even with a large ratio if missing views, multi-view models still significantly outperform single-view models. With 50% of views missing, the multi-view model trained with missing views achieves 57.04% top-1 accuracy, 16.39% higher than the single-view model's 40.60%. Addi-

Table 6. **The Top-1 accuracy of I3D and VTN models.** The rows indicate the vocabulary size of the training dataset, while the columns represent that of the test set.

	Multi-VSL 200		**Multi-VSL 400**		**Multi-VSL 1000**	
	I3D	*VTNPF*	*I3D*	*VTNPF*	*I3D*	*VTNPF*
Multi-VSL 200	66.35±0.0066	87.99±0.0051	-	-	-	-
Multi-VSL 400	62.3±0.0089	84.6±0.0029	63.14±0.0039	86.62±0.0045	-	-
Multi-VSL 1000	64.65±0.0137	72.49±0.0084	59.92±0.0058	77.79±0.0055	60.35±0.0055	76.32±0.0034

Table 7. **Performance of three-view I3D under varying missing rates during training and testing.** The terms *Full training* and *Partial training* refer to models trained with complete views and with missing views, respectively. The ratios in the *Missing rate* column represent the proportion of data lacking one view. The final row presents the result of the I3D model trained on single-view for comparison.

Missing Rate	**Top-1 Accuracy**		**Top-5 Accuracy**	
	Full training	*Partial training*	*Full training*	*Partial training*
0%	60.35 ± 0.0055		86.52 ± 0.0021	
10%	55.45 ± 0.0019	59.17 ± 0.0020	78.02 ± 0.0022	86.47 ± 0.0023
25%	47.78 ± 0.0043	58.17 ± 0.0029	75.74 ± 0.0028	85.77 ± 0.0026
50%	43.12 ± 0.0047	57.04 ± 0.0050	68.43 ± 0.0051	85.79 ± 0.0031
Single-view	40.60 ± 0.0040		73.03 ± 0.0046	

tionally, it achieves 85.79% top-5 accuracy, a 12.76% gain over the single-view model's 73.03%.

4.5. Effectiveness towards ambiguous cases

In the following, we employ I3D as the baseline model and compare the performance of models utilizing one-view and three-view approaches in handling ambiguous gloss pairs. Table 8 presents the most frequently confused gloss pairs for the I3D one-view model, along with their corresponding confusion counts in the three-view model. These glosses often exhibit similarities in hand gestures, and some confusion arises from related semantic meanings (*grain* and *desert* both relate to land and cultivation; *dry* describes land conditions, while *graze* refers to animals feeding, usually on dry or grassy land). It is evident that the errors between these gloss pairs have been reduced with the three-view model. The confusion ratio of all most confused pairs has been improved with the three-view, with the reduction in classification error from 12.50 % to 55.56 %, and some confusion pairs were reduced to 0 (*papers-blanket*, *cheer-rag*). This highlights the benefit of multi-view data in distinguishing challenging gloss pairs by enhancing the model with visual data from diverse viewpoints.

Table 8. **Most confused gloss pairs by the I3D One-view model**, and the confusion statistics of those pairs for the three-view model. The meaning of the glosses are translated to English.

Confused Pairs	**Confusion Ratio (%)**		
	One-view	**Three-view**	*Imprv.*
dry, graze	77.78	44.44	*33.33*
grain, desert	66.67	22.22	*44.44*
paste, huge	75.00	37.50	*37.50*
cover, reserve	66.67	11.11	*55.56*
papers, blanket	55.56	0.00	*55.56*
pick up food, weekend	62.50	50.00	*12.50*
film, edit	55.56	11.11	*44.44*
cheer, rag	55.56	0.00	*55.56*

5. Conclusion

In this paper, we introduced a large-scale, multi-view SLR dataset, featuring three camera angles and a vocabulary of 1,000 sign language glosses performed by 30 signers. To our knowledge, this is the first large-scale sign language recognition dataset incorporating multi-view data. Our work involved extensive data collection, a double-check annotation process, and a Deep Learning-based preprocessing and de-identification pipeline. We conducted experiments using popular baselines for both single-view and multi-view data. Results showed that multi-view data improved predictive performance by up to 19.75% compared to single-view models. We also analyzed the effects of vocabulary size and missing views, revealing the complexity of large-scale sign language recognition and the negative impact of missing views during inference. Our findings indicated training with missing views mitigated this issue. To advance our work, we aim to develop deep learning models for better integration and fusion of multi-view data or to enhance single-view recognition using multi-view models.

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